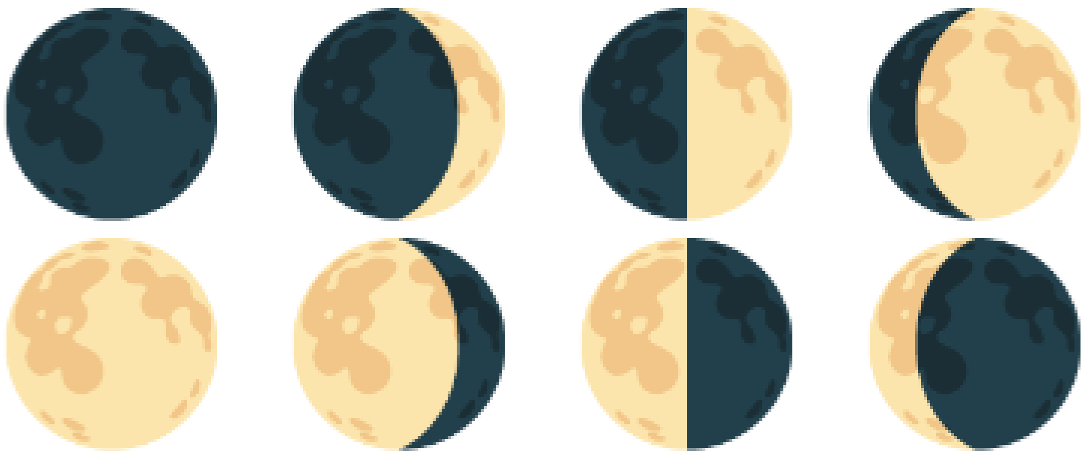
Little Lab Coats

🌙 The Moon — Earth's Natural Satellite

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 2 of 6

📚 **Grade:** 1st Grade
⏱️ **Duration:** 45 minutes + ongoing nightly observations
🔬 **Subject:** Earth Science — Sun, Moon & Stars
🏠 **Materials:** Household only



The moon moves through a full cycle of phases every 29.5 days

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* and *record* the moon's apparent shape over several days
- *Describe* the pattern of moon phases and predict what comes next
- *Explain* why the moon appears to change shape (we see different portions of the lit side)

🔬 The Big Science Idea

The moon seems to change shape every night — but it doesn't actually change! The moon is always a sphere (a ball shape). What changes is *how much of the lit side we can see* from Earth.

The moon orbits (travels around) Earth once every 29.5 days. The sun always lights up one half of the moon — but as the moon moves around Earth, our viewing angle changes. Sometimes we see most of the lit half (full moon), sometimes we see just a sliver (crescent), and sometimes the lit side faces entirely away from us (new moon). All the shapes in between are the different phases.

This creates a predictable, repeating pattern — one of the most reliable patterns in all of nature!

💡 **Key vocabulary:** moon, phase, orbit, satellite, reflect, crescent, quarter, gibbous, full moon, new moon, cycle

🌑
New Moon
Not visible — lit side faces away from Earth

🌒
Crescent
A thin curved sliver of light

🌓
First Quarter
Half of the moon lit up

🌔
Gibbous
More than half lit up

🌕
Full Moon
Whole lit side faces Earth — brightest!

🌖
Gibbous
Still more than half lit, fading

🌗
Last Quarter
Half lit again — other half this time

🌘
Crescent
Thin sliver before new moon returns

📦 Materials

🏠 What You'll Need

- This Moon Journal (printed — Student Worksheet page)
- Crayons or pencil for drawing moon shapes
- A clear night sky — or a moon phase calendar/chart (if sky is cloudy)

Timing tip: Starting this journal on or just after a new moon gives students the most dramatic arc — they'll watch the moon grow from invisible to full over two weeks. Check a moon phase calendar at ***timeanddate.com*** *to plan your start date.*

🔬 The Moon Journal Activity

Step 1 — Introduce the Moon (Day 1, in class)
Look at the moon phase diagram together. Ask: *"Has anyone ever noticed the moon looks different on different nights?"* Show the eight phases and introduce their names. Explain: the moon isn't actually changing shape — we're just seeing different amounts of the lit half.

Step 2 — Moon Model (Day 1, in class)
Take a flashlight and a tennis ball into a dark room. Shine the flashlight on the ball and slowly walk in a circle around it. Watch how different amounts of the lit side face you at different points. Explain: *"This is how moon phases work. The moon is always lit by the sun, but from Earth we see different amounts of the lit half."*

Step 3 — Start the Journal (Nights 1–14)
Each night before bedtime, go outside (or look out a window) and find the moon. Draw what you see in that night's box on the Moon Journal. If the moon isn't visible yet, write "not up yet" or "cloudy." If there's a new moon, draw a blank circle and write "new moon."

Step 4 — Name the Phase
After drawing, look at the moon phase chart and write the name of the phase next to your drawing. Does it match what you drew?

Step 5 — Mid-Point Check-In (Day 7)
After a week of observations, look at your journal so far. Ask: *"Is the moon getting bigger (waxing) or getting smaller (waning)?"* Make a prediction: what will it look like in 7 more days?

Step 6 — Final Reflection (Day 14+)
Review the complete journal. Fill in the reflection questions on the worksheet. Can you now predict what phase comes next?

💡 **Can't see the moon?** Cloudy nights happen! Keep a note ("cloudy" or "foggy") rather than skipping the entry. Comparing what you expected vs. what you actually saw is part of real scientific observation.

💬 Discussion Questions

🗣️ **Talk about these together:**

- What do you notice about the pattern? Does the moon follow a predictable order?
- If the moon is a crescent tonight, what will it look like in about two weeks?
- Why can we sometimes see the moon during the day?
- Has anyone ever noticed the moon looks bigger near the horizon? (This is the "moon illusion" — a trick our brains play!)

🌟 **Big Question: "Does the moon make its own light?"**

No — the moon has no light of its own. It reflects sunlight, like a mirror. That's why it's only visible when sunlight hits its surface and bounces toward Earth. The "dark side of the moon" isn't actually dark — it receives just as much sunlight as the near side. It's just always facing away from Earth. When we say "dark side" we mean the side we never see, not a permanently dark side.

🚀 Extensions

- 📅 **Full Cycle:** If you started with a new moon, continue the journal for a full 29-day cycle and see the moon return to new.
- 📚 **Read Aloud:** *"The Moon Book"* by Gail Gibbons or *"Papa, Please Get the Moon for Me"* by Eric Carle
- 🌍 **Fun Fact:** The moon is slowly moving away from Earth at about 3.8 centimeters per year — about the rate your fingernails grow!

👩‍🏫 Parent/Teacher Notes

- **This is a long-form observation activity:** The Moon Journal works best over 7–14 days. Build it into the bedtime routine — kids love having a reason to look at the sky each night.
- **Moon visibility:** The moon isn't always visible at bedtime. Around new moon, it sets soon after the sun and won't be visible in the evening sky. Check a moon phase app so you know what to expect.
- **Common misconception:** Many children think a lunar eclipse causes the phases. Clarify: phases happen every month because of the moon's orbit; eclipses are rare events when Earth's shadow falls on the moon.
- **The "waxing/waning" vocabulary:** Waxing = growing (getting more lit up). Waning = shrinking (getting less lit up). These are real astronomy terms and first-graders can absolutely learn them.
- **If cloudy for most of the 2 weeks:** Use a monthly moon phase chart or an app like "Moon" to show what the moon looks like — then have the child draw from the chart instead of direct observation.

🌙 Student Worksheet — Moon Journal

Unit 1, Lesson 2 — The Moon

Name: ________________________________ Start Date: ________________

Each night, go look at the moon and draw what you see in that day's box. Write the phase name below your drawing!

Night 1 Date: ______ Phase: _____	**Night 2** Date: ______ Phase: _____	**Night 3** Date: ______ Phase: _____	**Night 4** Date: ______ Phase: _____	**Night 5** Date: ______ Phase: _____	**Night 6** Date: ______ Phase: _____	**Night 7** Date: ______ Phase: _____
Night 8 Date: ______ Phase: _____	**Night 9** Date: ______ Phase: _____	**Night 10** Date: ______ Phase: _____	**Night 11** Date: ______ Phase: _____	**Night 12** Date: ______ Phase: _____	**Night 13** Date: ______ Phase: _____	**Night 14** Date: ______ Phase: _____

✏️ Reflection Questions

1. Did the moon follow a pattern? (circle) **YES** / **NO** — Describe what you noticed: ____________________________

2. The moon got (circle one): **bigger then smaller** / **smaller then bigger** / **stayed the same**

3. Does the moon make its own light? (circle) **YES** / **NO** — Where does moonlight come from? ________________

4. Draw what you predict the moon will look like in 2 weeks:

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

🌙 Moon Journal — What to Expect

The specific phases your child observes will depend on when they start. The general cycle: new moon (not visible) → waxing crescent → first quarter (half lit) → waxing gibbous → full moon → waning gibbous → last quarter → waning crescent → new moon again. The full cycle takes 29.5 days.

✏️ Reflection Answers

1. Yes, the moon does follow a pattern. It follows a predictable cycle of phases that repeats every 29.5 days.
2. The moon gets *bigger* (waxing — more lit each night) then *smaller* (waning — less lit each night). Then the cycle repeats.
3. No, the moon does not make its own light. Moonlight is sunlight reflecting off the moon's surface.
4. In 2 weeks: if starting from new moon → should be near full moon. If starting from full moon → should be near new moon (very thin or invisible).

💡 Moon Phase Quick Reference

- 🌑 **New Moon** — Moon is between Earth and Sun; lit side faces away. Not visible at night.
- 🌒🌘 **Crescent** — Less than half lit. Waxing crescent (right side lit) → growing. Waning crescent (left side lit) → shrinking.
- 🌓🌗 **Quarter** — Exactly half lit. First quarter is waxing; last quarter is waning.
- 🌔🌖 **Gibbous** — More than half lit. Waxing gibbous → growing toward full. Waning gibbous → shrinking from full.
- 🌕 **Full Moon** — Earth is between Sun and Moon; we see the fully lit side.

✅ What Success Looks Like

- Child can name at least 4 moon phases and put them in order
- Child understands the moon follows a repeating pattern (~29.5 days)
- Child can explain that the moon reflects sunlight rather than making its own light
- Child can predict what phase will come next given a current phase